Practical No.: 10

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COURSE: AI&DS
CLASS: BE
SUB: Computer Laboratory-IV

Title: Perform the data clustering algorithm using any Clustering algorithm.

0.1 Data Clustering Algorithm using K-Means Clustering

0.1.1 Step 1: Import Required Libraries

[18]:
```
import warnings
warnings.filterwarnings('ignore')
```

[19]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
```

0.1.2 Step 2: Load the Dataset

[20]:
```
# Load iris dataset
iris = sns.load_dataset("iris")

# Display first 5 rows
print(iris.head())
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

0.1.3 Step 3: Data Preprocessing

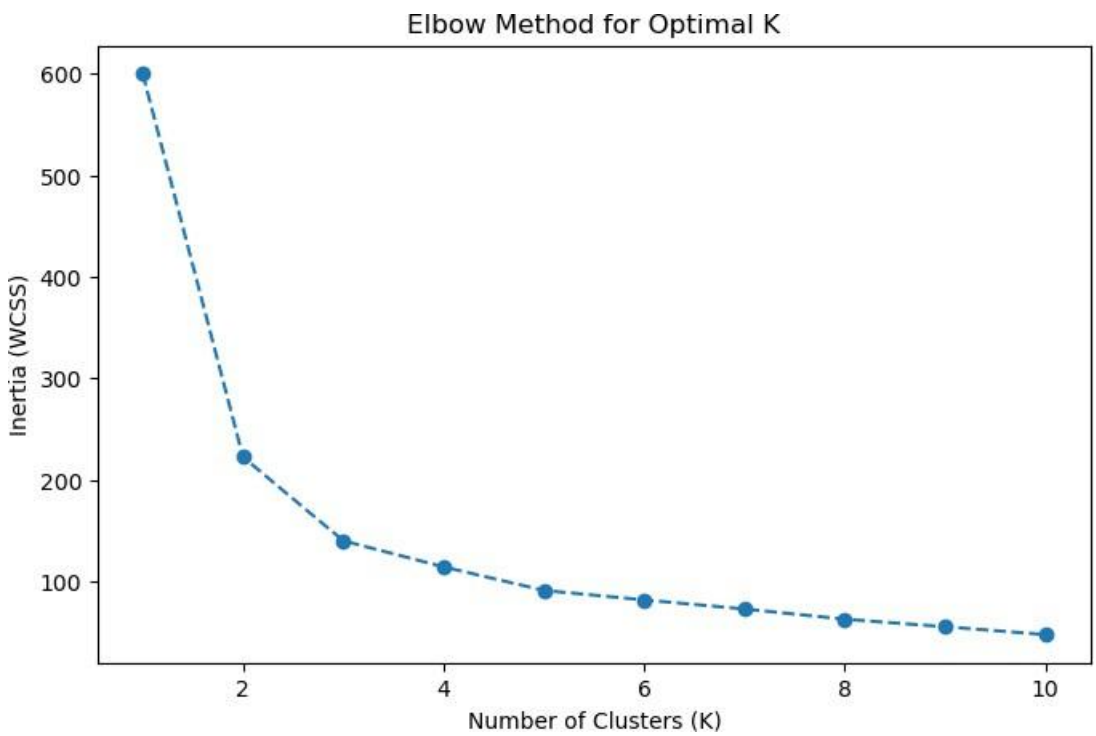
[21]:
```
# Drop species column
X = iris.drop(columns=['species'])

# Standardize the dataset
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)

print("Scaled Data Shape:", X_scaled.shape)
```

Scaled Data Shape: (150, 4)

0.1.4 Step 4: Determine Optimal Number of Clusters (Elbow Method)

[22]:
```
inertia = []

K_range = range(1, 11)

for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)

# Plot Elbow Graph
plt.figure(figsize=(8,5))
plt.plot(K_range, inertia, marker='o', linestyle='--')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia (WCSS)")
plt.title("Elbow Method for Optimal K")
plt.show()
```



0.1.5 Step 5: Train K-Means Model

[23]:
```
# Apply K-Means with 3 clusters
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)

clusters = kmeans.fit_predict(X_scaled)

# Add cluster column to dataset
iris['Cluster'] = clusters

print(iris.head())
```

	sepal_length	sepal_width	petal_length	petal_width	species	Cluster
0	5.1	3.5	1.4	0.2	setosa	1
1	4.9	3.0	1.4	0.2	setosa	1
2	4.7	3.2	1.3	0.2	setosa	1
3	4.6	3.1	1.5	0.2	setosa	1
4	5.0	3.6	1.4	0.2	setosa	1

0.1.6 Step 6: Evaluate Clustering Performance

[24]:
```
silhouette_avg = silhouette_score(X_scaled, clusters)

print("Silhouette Score:", silhouette_avg)
```

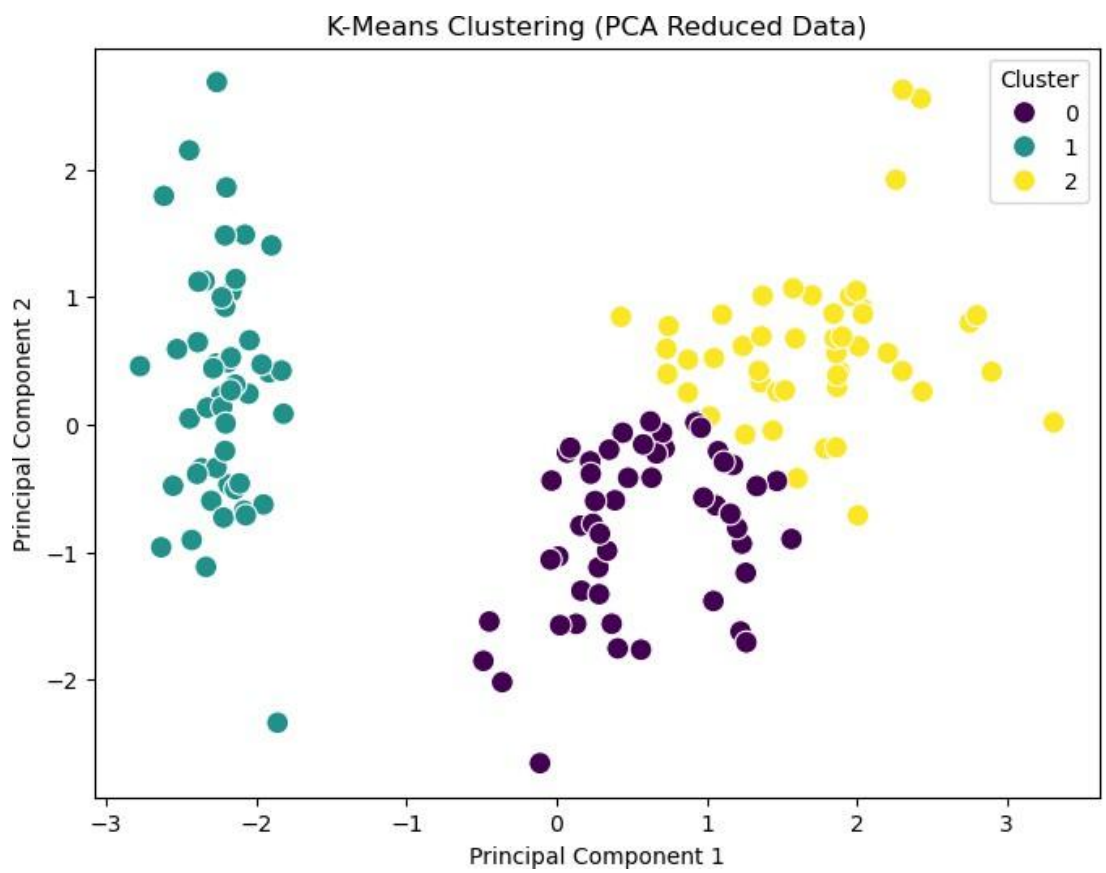
Silhouette Score: 0.45994823920518646

0.1.7 Step 7: Reduce Data to 2D Using PCA

[25]:
```
# Apply PCA
pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

# Create dataframe
pca_df = pd.DataFrame(X_pca, columns=['PC1','PC2'])

# Add cluster column
pca_df['Cluster'] = clusters

print(pca_df.head())
```

	PC1	PC2	Cluster
0	-2.264703	0.480027	1
1	-2.080961	-0.674134	1
2	-2.364229	-0.341908	1

```
3  -2.299384  -0.597395          1
4  -2.389842   0.646835          1
```

0.1.8 Step 8: Visualize Clusters

[26]:
```
plt.figure(figsize=(8,6))

sns.scatterplot(
    x=pca_df['PC1'],
    y=pca_df['PC2'],
    hue=pca_df['Cluster'],
    palette='viridis',
    s=100
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("K-Means Clustering (PCA Reduced Data)")
plt.legend(title="Cluster")

plt.show()
```